

ActInf GuestStream 044.1 ~ Tsuchiya & Saigo:

Category Theory, Consciousness, Integrated Information

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hello I'm higher to sign okay so just one second I'll just make sure it's going it's all good right welcome to both it is June 13th at zero UTC we're here in active guest stream number 44.1 with hayato saiko and nasugu suchia so really looking forward to this presentation and discussion hayato please um say hello and then we'll continue with nasuku's presentation hello I'm hayatussago I'm a mathematician and mathematical physicist also interested in Consciousness and a real working I work I have worked a lot with uh Now by using category Theory and application to IIT or other Consciousness Series so now all right so uh my name is now tutia I'm a professor in Monash University and also uh having a lab in ATR in Japan and other sort of the introduction to this today's session I'll give a brief talk on our recent project called the Korea structure which uh python I will be collaborating and also a little bit on the quantum Courier hypothesis uh both of them are hopefully relevant to active inference uh issue so uh the the quota structure project is the Japanese funding that we received recently and uh it's going to continue for five years uh this will be uh to establish a super interdisciplinary research program to tackle the program of Consciousness especially qualia through the interdisciplinary research and we are planning to also host a summer school for two to three weeks every year and here's the ah fantastic members for this uh Grand and hayato is uh basically uh working with me to uh uh in the part of the mathematical approach to the Consciousness and the Korea and uh uh so one of the program for example we want to address is for example when you see this uh Square uh Daniel what do you see here blue yeah so you say blue and I say maybe AO in Japanese and the height of may say Russian uh mean the same experience Gloria but how do we know that we are actually experiencing the same thing and I I'm actually curious whether this kind of question is something that free energy or active inference kind of style approach can ever tackle that my Approach is to go into the category Theory kind of inspired uh Direction and here's the first study that published out uh

you know we publishing a few years ago with ayato and the Shigeru together with also the another category theorist Steve Phillips that in category Theory there's an interesting Lemma called the unitarama that proves that all the relationship characterize its internal intrinsic property and we are using that idea to study The Courier that's the kind of an inspiration of our approach so instead of you know trying to characterize what the Brew is or red is we are trying to um characterize all the relationship between a particular color or particular Korea to other exorcery and then by finding the correspondence of this relationship we try to identify uh what the actual blue is so if you will say blue is similar to Green but and also similar to Red then that's going to be very different from my group and that's the way how we you know detect this you know blueness is same or different and uh one recent application of this kind of idea is this uh unsupervised alignment uh that we are using with another collaborative muscle for me always mean University Tokyo and I'll skip the detail but basically what we are doing is to do some kind of experiment and then collect a lot of data to obtain uh so-called similarity Matrix between all the colors and then uh Without Really knowing what color maps onto what we try to use on the relationship to predict what we are seeing so in a sense this is a purely category theoretical kind of you know approach to uh the program of consciousness so then I'll just skip uh the details here and then uh uh yeah so we are doing this kind of approach on other types of Aquaria and then this is the thing that we are planning to do for Wow next five years and then that another just brief kind of you know introduction if you have some other uh question uh we can come back later so this is potentially more to do with that active inference aspect of our approach in a sense and this is called the quantum Courier hypothesis uh we we published this uh preprint really recently and so you can access it later so um one of the uh there are many programs in the current you know uh models of the quality or Consciousness where we tend to think that um experience or Consciousness is like you know one point in the high dimensional space and we use that kind of idea for our previous you know alignment experiment uh approach but uh it lacks Dynamics and it also lacks the you know effects that we uh tend to experience where you know the observation or report or introspection itself sometimes change our quality and um that's the sort of the active inference aspect that I'm talking about you know um attention or introspection changes about experience but the traditional approach is very difficult to deal with this and one way to deal with that is this uh uh Quantum formalism that we think uh is appropriate so here we think that you know there is some kind of Gloria experiencing system that is defined as this blue boundary you can consider that as sort of the mark of blanket like thing in the free energy kind of idea and then

we consider a particular type of the quail like redness a balloon as itself as an observable it's a terminology in quantum theory uh something that in principle can be observed within this Warrior and then we consider this measurement as a process that probe this accordions um you know observable and unlike a classical uh favorite theoretical thinking we think that the measurement or probing of this observable through it's called a state changes its you know States and also observable itself and that's something that we can actually formalize mathematically using a concept called a very general concept of this uh conditional probability called the instruments it's a it's very technical for the you know uh quick talk so I'll leave it but you know this is a modern Quantum uh theoretical idea that captures both you know how we can uh affect on the world and the the quota itself can be affected by that measurement and then using this we think that you know um uh we can explain properly Korea in a much more reasonable way with the mathematics and uh solver it's just a kind of theory but we already think that it's probably better than the existing kind of a model of the for example attention and Consciousness and also uh it also uh provides some kind of you know prediction and the explanation of why some of the two concepts or two experience can feel different uh depending on the order that you experience so this is just like conceptual kind of example but if I ask you Danielle that uh is China similar to North Korea then you might say probably no right I I don't even know the reference point yeah okay but let's say you know just objectively you know China doesn't feel similar to North Korea okay yeah however if I reverse the order it's not North Korea similar to China then many people actually start to say that ah yeah it's kind of similar does that have to do with one being larger or what is it about the order that primes people to respond differently yeah so many people have already you know come up came up with many different kind of ideas to to explain this kind of individual aspects of this you know older effects but as we uh mentioned you know we are actually Computing or you know measuring this kind of order effects systematically through all combinations of the colors for example there it's hard to systematically explain why this other effect emerges in One cases but not the other and in which direction right but well that's what we found basically and but using this in a Quantum cognition kind of framework we can explain it really nicely in this case it's just a projection post rate but uh we are probably going to extend much more in the more instrument like you know a framework in the future yeah so and also we can we are also predicting that there may be some kind of well uh violation like uh a violation of value inequality like situation in uh you know uh perception and if we can confirm that that also you know gives a Credence to our proposal that the quality is like you know it needs to be

uh framed or explained in the quantum framework so I think that's probably enough to uh start up the conversation I guess awesome okay great lot to uh jump into all uh share my screen so we can take some notes um how so do you want to give a first remark or I'm happy to but but do you have a first remark or what parts of the work do you want to reflect on first foreign [Music] I think maybe the categorical approach the importance of the categorical approach is one issue to talk uh uh with you I think because uh you first pointed out that uh paper uh with Shigeru taguchi in 2016. um so I'm uh interested in why you and your colleague are interested in that paper from the active inference side so can you explain some feeling or foreign yes so this is um using category Theory to assess the relationship between Consciousness and integrated information Theory with the two of you and taguchi from 2016. so I've long been interested in scale free or scale independent ways to describe systems and information theory has been recognized as one of those approaches and information theory has been proposed to not just play of course a central role in the functionality of information systems but with IIT and related theorems to also strike at the heart of the most quintessential aspect of our conscious experience so that seems like an incredible synthesis of the functionality of systems that don't necessarily seem to have anything quality alike at all merely performing calculations like a super fast Abacus basically um and so I am curious and feel free to address either of these things when we describe Information Systems what is it that would or could or is uh experienced are they experiencing their the their information what they are processing or what is it about information integration would Grant a system the ability to move Beyond a behavioral functionality into again strike at the heart of one of the most ineffable questions that the mind can really consider okay interesting um actually the category itself uh in mathematics originated are created in mid-20s centuries uh to bridge the two different fields that is uh geometry and algebra uh more specifically the topology you know something about what was you maybe um which is a very uh General kind of uh mathematical methods to capture the essence of the figures uh for like a circle or the complicated uh figures and the the fundamental program or the topology is you can you identify the two different uh figures essentially that means you can can you transform continuously to the other figures or not so if the one figure goes to other figure with A continuous transformation and back then it is uh viewed as essentially the same but the two the screen to to determine the two figures there's chances are the same is a very difficult question so um like great mathematician right figured out the utility of the algebraic methods that like making the one figure to the other uh from the figure you can construct the algebraic system like groups or something the uh the transform uh

the problem into the algebraic region the then the point is very essential point is uh taking one figure to the uh the uh constructing one algebra from the one figure is not sufficient the point is the deviation should be transformed the the other regions so one region here other regions here but one two one correspondence of object is not surface entry but the relations if you can transform the older relations into the other regions then you can uh provide uh you can prove something from one region to another region because uh yeah this is a notion of the fact uh like the region intuitively I talked about the term using term the region it is mathematically called category category it consists of objects and arrows arrows or amorphism which is a generic kind of relationship between these objects so our topological region is some kind of category of topology like a top that objects the topological space and the continuous functions continuous mapping is a role or something the other regions we can for example as a recreation like groups and group homomorphism or the module Vector space or something so something RJ rack and the relationship between them so category is uh defined to formulate the uh intuitive concept of bridging something function so first the Arenberg and Mark brain defined category to Define this notion of function by the bridging to region then if this is established then you can infer about the topological reasons category by using the algebraic the relation or something so we uh that is now on shiger and me uh talked about the possible application of this kind of argument into the um uh conscious and study that it is just a problem of bridging different kind of regions like uh physically uh testable religion under our role experience itself or something so it is apollia it seems appear it is a two difficult question too hard problem to connect to these decisions but if you only focus on the object but the relationship if you can compare the relationship then you can infer something about you cannot directly touch you directly manipulate so this is the idea yeah maybe I'll just add a little bit um so uh yeah as I had said oh you know under you you are taking really diligent notes so that may be uh quite good but uh I from my end uh so this uh paper was inspired by really talking to Shigeru and the meeting in this summer school on Consciousness and there I met I encountered in this in a category theory for the first time and uh there were several shocking kind of you know uh Concepts that I learned from this link first encounter one is that you know there isn't a way to compare two completely different things and then extract the common structure between them at a higher to first introduced me this you know very strong strange paper which I didn't understand at all but was so interesting that I could read you know for hours and hours it's called the Rosetta Stone it points out uh commonality between the topology quantum physics logic and computation yeah yeah that was the title and then I didn't

understand what the heck is this in a paper but uh that's that uses this category Theory and uh as a whole I feel like ah I see so maybe if we understand the structure behind this each of this a concrete example we may be able to find a commonality or explanation at that level and that naturally applied to my you know program at the time which is you know how to bridge the gap between the Korea and the information structure derived from neural you know uh activity this is this looks also completely different but if there is any possibility to bridge the gap it must be some kind of structure or mathematics and that's this category Theory hello one point is uh one other point is that you know you I I I think you uh you can get the ideas and functions by breathing the two regions then the really interesting thing another really interesting concept is called natural transformation natural transformation is the uh kind of deformation or transformation between transformation so you imagine some Theory Bridges some addition to the another region so there is a one meta level Arrow code but uh another theory uh predicts another thing or something but can you connect these theories or models this is a problem of natural transformation so one representation one Theory or something uh uh May map the one region to another region but there is a there may be another various kind of mappings functors so the problem is to compare or to unite to connect this transformation is a functions is called natural transmission if the function is naturally isomorphic that means there is a reversible natural transformation between functors then uh these Essence these series seems different but essentially the same so kind of natural transformation is maybe the hint to connect to the various series on Consciousness or something and also the our Consciousness experiences itself will be connected to the concept of natural transformation because the I see something that the different thing something but there is no uh divided discrete kind of construction but there are some continuous or relating something so this is this view view is something that like function the another view is something like function but there is no division there is some relationship so our experience maybe uh itself a kind of product of uh natural transformation like yeah maybe now can you you can explain like a peripheral oh maybe ah no you're muted muted is it all right if I draw some kind of figure oh of course would you like to share your screen yeah and also if you can give me the access to you know uh to The annotation um I will um make it so everyone can edit and I'll put it into the chat here and then also if you want to draw that would be awesome yeah thank you now looks good yeah so just to kind of recap what the Hyatt was talking about is that uh we we tend to think about something like a category and then another uh category is there and then there is some kind of object which which could be quite uh you know abstract and then it's related by Arrow

like this and then if this category C is uh somehow possible to map onto this category D while kind of preserving this you know um arrow for example this arrow goes to here and this arrow goes to here and then each object is also you know preserved then it's much more than the usual kind of idea of a function between the sets right because it also impose the you know correspondence between the relationship and in our idea basically if this is let's say G of the category and if this is a neural uh activity or uh maybe information domain and then maybe you know we can have a background force of the functor that would be great usually uh the most of the mathematical Theory Of Consciousness tries to go from here to here right let's say in the case of IIT Arrow here and then what what Hyatt was kind of talking uh in a way as a natural transformation is uh this kind of you know secondary level Arrow let's say I don't know whether free energy or active inference can uh count as a prediction from you know neural activity to the particular query but let's say if there is something like that and then maybe there is some kind of a translation between these two theories yeah if that exists that's called a natural transformation awesome well a few comments on this the idea of mapping the maps between the spaces is so important it reminds me of metaphor where we're conveying relationships and saying well it's it's like a hammer or it is a hammer for this other situation so that's how we get few shot learning and applicability and how we grasp Concepts is perhaps developmentally or from an evolutionary perspective in a large extent by generalizing our spatial and embodied understandings and then certainly communication if you want to have some semantics that are being communicated across Minds through an interface through a classical interface then certainly you want the structure to be preserved and then I think the one that you have here is one of the most provocative for those empirically studying uh consciousness because it's kind of like this side is empirical and let's just say we don't know what the Q side is for now but let's just say it's another space that isn't that if there is a certain uh landscape that neural activity patterns flow through like some kind of winterless competition or you get some sort of limit cycle and you find that there's a limit cycle of neural activity and then there's also uh that is being paralleled by experiential Cycles then that is not just an accounting system that's a very strong finding about potentially the real underlying causal system especially because it can suggest unique other explanations or manipulations to make do you want to mention anything about the metaphor before that uh very hard to say the technical remark is that the for the original uh problem that connects topology and algebra then in that case the algebra is more concrete more manipulatable in that case and the topology is something mysterious in this case so in general the function is a kind of um generation the

mapping in sets so the something mysterious into something concrete so now the picture C and D should be switched maybe in that case and yeah but uh of course there is a background maybe back and forth so the idea is this our you are brew and my Brew program is for example you can prove that you can uh your blue and my Brew is different if there's something uh informational or something the detectable structure is completely different they are you brew and micro is different of course you cannot uh uh prove this the sameness between you your blue and uh my Brew uh if there is no how to say the back and force function but even if you don't have the uh the backwards function at least you can demonstrate that your blue and my Brew is different seems to be different if the correspondence is factorial or something so it is a big step from The Experience answer site that and with me and colleagues uh do such a thing so you have my my your but the this is not uh by focusing you approve my Brew only but your blue read something or you don't have to say even do not have to say the name of something so just uh how to say no labeling for example but the relations you you can teach the relationship at these things you can teach the relationship with these things so then compare that so very interestingly uh in general the structure is surprisingly uh uh coherent uh between persons proper persons or yeah so it is very nice and also maybe it's it's suggesting that the non-isomorphic case then you are blue and my Brew is something different or something so I think this approach is very interesting even if um in the no bi-directional Only One Direction then but even that case it is useful yeah it is my remark yeah please please and then I'll have a question about your drawing right so this is a one of the slides that I didn't show today but uh uh what what hayat was talking about is um you know kind of you know explaining this figure like uh the way you experience red is related to with the other experience uh you know not only color but shape and also with sound and so on in a very very complicated way and then we can actually measure how similar they are by asking the people and then you can also do that for the other people and then these these two similarity Matrix can be directly compared with our you know new method of this unsupervised learning and also IIT for example may allow us in in the future to go back and forth between these you know uh scoria structure and the information structure and then mathematically we believe that the information structure the information structure should be potential you know translatable in some way and if that is possible in some way you know eventually we may be able to say ah okay so the way you are experiencing red same or different in this way or that way that that's a sort of the idea can you go back to your drawing the top Arrow as if we're kind of just exploring what what might it mean that IIT is going back

and forth and also that active inference is going back and forth so what exactly does the edge between C and D represent in IIT is that a quantitative evaluation like the five variable or what is being accounted for with that Arrow mm-hmm do you want to say something yeah uh there is a various possibilities actually for example the sameness was different or so if you have some kind of gradation with sameness so yeah same as gradation but to a former to to formulate The Graduate gradation gradation of uh sameness you have to move on the air reached category not a simple kind of category but it's a generalized kind of category called enriched category in that case for example the simplest example of enriched category is a generalized General sorry generalized Matrix space metric but the like uh in Matrix space for example the uh the point p and p itself has a distance zero of course and uh kind of P to q q to R uh is uh there is a triangular inequality or something so yeah then it can be considered as a generalized kind of uh compositional structure called enriched category it is a typical kind of in this category uh so but the symmetry does not uh always hold or something so this is a one how to say the uh one small step to uh grasp the our Courier structure because uh like uh now pointed out North Korea China distance and the China North Korea distance is uh in general different or something so almost all studies uh assume that some symmetry but we need to go go uh beyond that maybe the one number or something is too poor to represent everything so more you have to go more and more richer structure like not just numbers but some kind of projection operator or something then you enter into the quantum region relationships we feel should we maybe should be uh represented from the some kind of the relationship between the notion of State in kind of quantum Theory under State transition between that so our approach is step by step and you can choose the many arrows to study something under something and then goes to unify the more and more General kind of arrows and categories and I this is just for the speculation but this kind of quantum theory theoretical like other mathematical framework and not do not need to that kind of physics but they're called mathematics can you use unify these kind of things awesome but just a few comments on on that when we're doing free energy minimization we use the KL Divergence which is very importantly not a distance because a distance would have the Symmetry characteristic and so the KL is chosen because it's easy to fit uh optimization on these variational distributions but it's almost like it's a feature not a bug that the KL Divergence between q and P is not the same as p and Q because it turns out that that kind of a of a divergence-based um uh quantification actually has some advantages um over a distance measure and then the mapping between observables and hidden States and the cognitive operators that we bring to bear on that kind of mapping problem those

are some of the most fundamental pieces of the partially observable Markov decision processes that are used in the active inference generative model which is first the States and then how do hidden States change through time and how does Action selection policy selection influence how hidden States unfold Through Time so this mapping between hidden States and observables in between hidden States and the hidden State at the next time points are and how again how action influences if you understand those operations then it's like a game of chess and the ball just rolls downhill and the variational distribution Works exceedingly well but when those are off base then you're fitting your way like right off a cliff so uh maybe again I'll just add one uh more points so you you asked uh in the original like five minutes ago wow what this you know each of these points or Arrow correspond to in the case of the IIT uh and so on and then this is kind of uh related to all these uh ideas but uh well one of the um you know fundamental uh insights uh from IIT but also from philosophy especially uh some kind of philosophy called the process theory is uh it is uh important to really understand that the process of law or movement is really fundamentally um important kind of aspects of things and then um that may be really distinguishing some kind of static set or um the thing uh in the static world and when they start to interact and then when the process emerged then something uh starts to potentially you know uh generate or support subjective quality like you know uh protocol kind of thing and in the case of the IID uh it is uh deeply kind of you know uh entrenched in this you know formation of that transition probability Matrix of an entity like how this thing neuron or logic gate it might have come from the previous state and it may go into the next state so any kind of a moment is already embedding the possible previous and the possible future there is some kind of a with us in a moment because it's a process and when the process is considered then you know as I have said that you know this kind of you know dot is a very you know kind of you know two Thin kind of structure to think about so we need more elaborated structure of a DOT itself but the category theory is great uh in a sense that each dot can itself contain complicated structure it's just you know wow uh represented as a single dot but it can't contain something more right so the in an upshot one dot could correspond to a collection of the neuronal you know States and the connectivity pattern and things like that which can go from one to the other and then that may that this kind of movement May correspond to something like you know we see redness or whatever in this way but that's also with respect to our other kind of you know arrows as well and that that's the thought that the wave of the relationship that we are trying to characterize both in the domain but Aquarius but also brain is also the wave of the relationship gigantic

ones right and as you said that you know transition from one state to the another state that's pretty much like dynamical State uh that dynamical system like you know approach and then uh the simple idea called a monoid is uh applicable to this kind of dynamical system approach uh correct me higher though if I'm saying something you know out of the bounds of my comfort zone but uh you know any kind of a state can transition into another state and then this tradition itself can be represented by the arrow and uh yeah so uh and that kind of the combination of that particular kind of Arrow composition May correspond to one so that that's uh uh one thing I wanted to say the other thing that I had also one uh it was then mentioned about Observer brand states and why we are interested in the quantum uh you know framework quantum theory by the way it's not a Quantum brain hypothesis we are not saying that you know Quantum you know phenomena in the brain is critical for Consciousness but it's just a mathematical structure of the quantum theory that is important to understand the Korea and the essence is most likely to be this you know non-commutativity any kind of process that you do like you know cooking you know putting the water and then uh model is different from model and then putting water right and so any kind of process we do there's some kind of you know temporal dependency the commutativity doesn't hold and that's also probably the case for the Korea and if that's the case uh then for you maybe it's a bit surprising but you know Bayesian kind of statistics and Beijing probability may not work as you might wish yes so the in other words using the counter mathematics the there is a quite good generalization of Bayesian uh Innovation or Vision kind of uh inference yes uh actually the the the the one step generalization is called the projection uh in kind of mathematics that are then there is more generalization code uh instruments so instrument is a general notion of measurements like a system and the environments uh interacting the then more precisely the measurement is something that like the interaction feature has some probe over their uh apparatus so probe is some kind of the the the apparatus is some very small tips or something which can be considered as a Quantum that this Quantum system and this probe system interacting randomly then we can measure this probe by the macro side or the apparatus so this is a very general kind of uh measurement uh which is called which is identified mathematical with the notion over uh instrument originated by the Davis and Lewis and the uzalar one of our collaborators we begin in the cooperation but other uh who is also famous for the ozawas in inequality although inequality as a generalization of the Heisenberg University version of heist making important but uh he uh identified that this General notion measurement and instrument called completely positive in instruments in mathematics but this CP completely positive

instruments is a quite General and generalizing the very beautifully the Bayesian influence into the quantum side so I think the the using and focusing this kind of instrument kind of concept is the key maybe the three uh to bridge the um active Independence side on our side and Quantum side or something so yeah it is my speculation but I think it is very fruitful Direction oh it's awesome just one comment on that the the probe with the instrument or the quantum reference frame I believe as Chris Fields might describe it is an experimenter's choice and so the way that we could talk about policy selection and inference on which experiment to run would be in terms of which sequence of observations is going to minimize expected free energy okay so to have high pragmatic value and epistemic value that's the expected free energy functional now as researchers we might seek out the most informative experiments and not prefer to see a certain measurement so we can say well all measurements are a priori equally pragmatically valuable to us so now expected free energy is going to be driven by information maximization just by pure learning but this recognition that action is woven in to the process of reducing uncertainty about other systems across blankets it's it's absolutely fundamental because if you don't have the constraint that you need to actively select which experiments to run and to probe and to interfere in this Quantum way this way described by Quantum mathematics you're kind of in a delusional alternative world where like you get information from photons but you don't have a sensor so it's not an empirically tractable World unless we take on this constraints and then once we do our mathematics are the quantum mathematics and again it's kind of a subtle point this isn't the orc or this isn't necessarily taking a stance one way or the other on what kinds of electronic scale Quantum phenomena exists in any system but rather just sees Quantum mathematics as fundamental to decision making and uncertainty which is quite a change from how it's been discussed for a hundred years actually yeah as now said we are not um depending on the hypothesis like uh there is a Quantum physical process Behind The Courier but the why we can uh use the quantum mathematics is very very uh simple reason because a conventional probability Theory practical probability Theory uh depend on the probability space classical one is too narrow to capture for example over effect or something but you can generalize it so kind of mathematics is just generalized the classical mathematics and I think I believe that the why Quantum mathematics is found for kind of physics is very simple reason because in classical physics the some determinism the classical determinism is a very Universal on the the probability only comes from the some kind of inner ignorance of us but in Quantum physical context there is some kind of very fundamental uh randomness of the fundamental uh accidents or something is found and they

are all surprised because if you go into the micro side the determinism should be stronger and stronger they thought but actually there needs some kind there they're found fundamental non-commutativity and macro micro side there was some people pointed out that Even In classical or evening sorry a microscopic side but uh this kind of uh the importance of non-commutativity is very Universal not especially in micro side but it is a historical reason that we first find in government mathematics in the micro size but there is many many phenomenas uh feeds cannot be captured by the determinism or the classical probability Theory so I think the Korea or something is just some kind yeah example yes uh by the way maybe Daniel can you uh replace this Quantum Consciousness with the quantum Claudia yes so the quantum concept has been used to you know uh implied this Quantum brain hypothesis which may be or you know potential you know uh true but the sulfur we don't really see the strong evidence about you know this um particular thing what's funny I I'm thinking of the classic qualia question what is it like to be a bat and it's like what if the like is interpreted as a metaphor what if there's a metaphorical mapping to bat which is an observer dependent mapping amongst some Spates you know among spaces hmm because we've moved the focus from the the objective Standalone experience of the bat to considering mappings amongst cognitive systems one critical one being for example whoever is listening to the question and a bat who's the one who actually answers that question oh okay so you you asked also about the you know the inspiration from the category Theory uh relevant to the uh issue of the metaphor right that was a communication and in fact um so now it's a bit you know far from this you know our Quantum query and they jump back to another topic but uh so the metaphor uh model of a metaphor is something that the hyatta has been working on so I'll yeah but the the other part which is this you know communication and that's something that we also wrote together uh last year um and using this you know concept uh called an adjunction in a category Theory which may be also interesting to uh discuss so 2016 paper we already mentioned that little uh category theory is a very promising for the study of Consciousness precisely because it provides different levels of uh sameness when I heard this from hayato I was like what what do you mean but you know as simultaneously it makes a lot of sense and in fact many many you know controversy or you know confusion among the you know field of Consciousness but not only for Consciousness other fields like ecology as well like you know uh classification of the animal species and things like that all up there depends on this you know what do you mean by the thing together right and this is uh something that also relates to what you are talking about a moment ago which is Observer dependence right so in the uh at

uh even at the really very important how it is you know our reality I think that what matters is some notion of this subjective choice of what it means to be the same equivalence or adjunction is a particularly interesting kind of situation where for example you know um integer and the real number is definitely different but in terms of the relationship if you start to think about the relationship there is some kind of you know a one-to-one correspondence between them and also uh Hiatus book also written in Japanese but uh recently I did it and one of the interesting example is also multiplication and exponentiation they you know at the level the object or number it doesn't you know match right but there is a very nice relationship between them that is very fundamental and something like this you know apparently completely different thing but finding really um meaningful relationship that's very fundamental in communication you know we are talking different kind of language with different kind of meaning but because when I say something you reply and then there is some kind of systematic difference between what I expect and what I hear that's the kind of prediction error right I think the prediction error may not need to be minimized in fact but as long as it's consistent and systematic then it actually works as a you know very nice translational kind of a device like I don't I I continue you know after living like 20 years more more than 20 years of living outside of Japan I still have very strong English accent either in Japanese or you know Japanese living in Australia or you know you get a us I can't minimize this prediction error but because I have this systematic prediction error for you you totally understand me right something like this is uh the uh interesting thing about that Junction well this this makes me think about uh Nest is reference frames in communication and let's just say that the the expected free energy or maybe even the integrated information content is such and such a level but it varies within some range it's 100 plus or minus 10. instead of just trying to naively minimize that to zero we can create a nested model such that we expect that to be 100 plus or minus 10. at which point the non-zero value is stabilized or buffered and so then that becomes kind of stabilizes or wraps the fact that we're not perfectly transmitting information but we can be perfectly aligned that we're not perfectly transmitting information and also our language understanding for fluent speakers were able to map the phonemes a little bit deeper we're able to read between the lines or listen between the phonemes so that people who say words map to the exact same syntax let alone the semantics the sound gets kind of abstracted out but it influences our qualia so something is being passed forward with like tonality and the timing and I wonder about with of course large language models how much context they're able to Garner from the string of tokens when for our experience uh sending and receiving

natural language communication it's all about the timing and the tone and the setting and so finding ways that we can describe that type of information the the category Theory capacity for the different levels of sameness seems like exactly the tool that we need I have to do you want to add anything yeah I'm thinking about but about that you know I have the multi-level kind of free energy I I actually have a question to you actually so I recently uh discussed uh with a roboticist in Japan uh June Tani yeah do you know his work yes yeah is uh using this predictive coordinating kind of scheme uh for um constructing the set of you know autonomous robot and uh he thinks this you know uh intermingo of the top down prediction and the bottom-up you know a prediction error is a place that you know Korea emerges and things like that but one one of the things you know most interesting uh aspect of that his model to me is that he has this different time scale for his uh neural system the most peripheral mortal m11 the sensory level has a very fast you know time scale so that tries match with you know what how what's happening outside and then uh uh what they want to do towards you know outside and the intermediate level has a slightly longer kind of you know time scale so there is some kind of inertia and an even higher level has a much much longer time scale so in DL probably neural system there is also different level of the time scale are the different levels of the mechanism are acting at the same time like um changing the activity state in our spiking or not and also released all the synapses uh uh synaptic transmitters that could increase the uh decrease over the time and also even the new neurons emerging and are going out so um the the minimization where you are talking about minimization probably that kind of minimization is actually happening independently and also coherently potential or contradictory across different levels another regions and there may be some kind of effort or whatever to also you know coordinate this kind of contradictory demands of the minimization we think about this issue do you still think it's an uh useful way to think about or use the term minimization in this situation yeah it's a great question well one that stood out to me was you described the slower moving levels as having inertia and from the Bayesian mechanics perspective that's exactly how we would model it they're slower moving they have more inertia and they move more like classical objects whereas the faster time scales are moving more like lighter particles more like that Quantum probe in the limit and then this sort of Center or the highest nested model would be like the most cognitively massive potentially the slowest potentially integrating information over the longest time window and certainly this is uh underlaying by these multiple time scales a mechanism spiking neurons from a few hundred milliseconds time scale neuromodulatory processes over seconds and minutes

synaptic plasticity even neurogenesis and are the development of our brain so what does it really mean that minimization is occurring when we have this heterarchy where we have similar types of free energy minimizing units or active inference units within a level like cortical columns or grid cell units and then also we have these nested models that are optimizing across uh spacious temporarily separated scales well one paper that looked at this was our livestream 42 on the slam simultaneous localization and mapping also in robotics and so they had a two uh nested model with two steps and the lower level was like the tactics and that was where you had the motor and the sensors and then the nested model was slower that was like where do I want to be that was more like the strategy and you can create an expected free energy functional that sums across those two levels how you tune the parameters is going to matter because how do you wait yeah yeah um like and one one this comes up with like preference learning like if you have if your preference if you have two left or turn and right turn and your preference Vector is like two and one versus two million and one million then that is the ratio of the preference is the same but the amplitude of the preferences is a lot different which is going to swing the model towards valuing pragmatic outcomes whereas if you had 0.02 and 0.01 the ratio of the preference would still be the same but it's such a small amplitude preference and so similarly with expected free energy minimization it's you could make it so that the The Joint minimization was driven by just the top level of the model and then this sort of contradictions or even increases in expected free energy of the lower sensory model would basically always be accepted in service of the top model or vice versa and it seems like biological systems with their sort of precarious or self-organized criticality operations they're poised where you can have like propagations where small changes in the bottom up model can propagate and um and induce like phase changes of higher levels and vice versa so they're certainly poised parametrically at a special place where the composition of the faster sensory motor models and the slower like cognitive or narrative or metacognitive levels it's all poised in a very integrated way potentially even one that is um where where the IIT measures tell us something useful but yes the the contradictory demands of subunits within a level of collective behavior and across levels as a biologist I think we just point to ecology Evolution and developments and just distributed systems that don't exist at an Adaptive point or Implement adaptive Dynamics amidst uncertainty they just fail to exist and so the complex biological systems that we observe today do manage that trade-off and then in robotics they have this unique challenge of developing systems with one shot or with iterated development that do have some of those properties but for sure there's no reason to think that just by laying

out a nested active inference model that you would actually get like a more adaptive decision-making agent in fact as you Nest models the state space these models becomes vast and with some back of the envelope calculations you can see like well if I do this level of nesting and have this kind of sensor array we need this absurd amount of data to actually parameterize this model and by the time we collected that many observations maybe the causal structure that we learn to the world you know 200 hours of video ago don't apply I mean that that could be a quite a huge difference between uh probably IIT and the free energy I I think um the IAD explicitory states that the decimal time scale that matters but in this in this case it's uh just about you know Korea is our particular time scale that maximize integrated information and that corresponds to the time scale of our experience you know model you know that um slowest uh time scale is like on the order of 100 seconds or something like that and also you know new synaptic plasticity is on the order of hours and uh most likely spiking neurons seems to be more mattering with respect to our particular experience at each moment right and the why it is the case um that the provisional answer provided by ID is at the because at that particular time scale integrated information is much more and across the scale and across the regions but um this is a huge conjecture and nobody knows whether this is true or not and uh when we use the fly model uh we weren't we uh we weren't able to sort of confirm any kind of a clear peak in the integrated information um using a different time scale so it's still you know something that I'm very interested in um looking but uh still not uh clear whether this is the case I don't know if hayato has um some um idea or um inside into the appropriate time scale for Consciousness or something from the physics point of view yeah yeah yeah I have some Souls that maybe that it becomes a long story so uh and also connects to the um Observer dependence uh problem so I I was thinking thinking about how to explain this very important things in one stress and I found that first um the very uh simple fact that there is no uh uh there comes no identically same two events uh every event is different in some respect but uh to do science or more General is some recognition of law we need some the Criterion of sameness for example you want to know how to feed at how to heal the disease uh you should identify the different persons people as the same same kind of uh the samples or something of course there is no identical persons but you should you have to uh the identify the the same in the members of the same group something so all the scientific studies is depend on some choice of the sameness and so in that sense all the science depends on this kind of choice or the sameness uh as you both pointed out and uh there is no I think that there is no canonical choice of the Samus you can choose every semeness but if you choose some sameness then uh

you can talk about The observables Observers means some kind of quantities you can observe or something maybe it's randomly change the value or no value like a count of case but by measuring process it it's become half the value or something so so they always may be indefinite uh nature of the such kind of quantities but anyway the the we can imagine that quantities has some kind of structure between the quantities so it in mathematical terms it it will be some the algebra of the observables then how to connect these ideas theoretical idea to the experimental uh thing is the key is expectation value so uh Theory provides expectation value of something and we can connect the theory and the experience experiments by comparing the theoretical expectation value and average of samples so all the quantitative science depends on this structure like algebra observous and some mapping from the observables algebra to the sum number system so this is called the state it is in mathematics and algebra and state is something like the mapping of quantities of random variables or observables into its expectation value so this mapping over expectation is technically called state so if you start start from Observer algebra and state this is a starting point of non-commutative probability theory that is if this algebra is commutative then the conventional probability is probability Theory can reconstruct from this commutative algebra and the state but if this is non-commutative as now pointed out if the process is essential then these algebra should be non-commutative then we have the uh mathematics or something so in this case The Observer dependence uh is described uh uh the the other kind of Observer dependence can be described because uh you don't you can't have the classical probability space for the old always always but if you choose one observables one observable like position or momentum you could you can choose this then you can construct the classical probability space so the choice of context or a choice of measuring something is by choosing something so context you can have the classical kind of reality so this Observer this kind of Observer with dependence is where formulated by using non-commutative probability space so I think the these multi-layered observable uh yeah uh Observer dependence is uh it can be described by using non-commutative probability other the last thing I would say is the connection of the category Theory unknown commutative probability uh very interestingly if the category category is given like a causal system called causal system give us some category like a causalization is Arrow and the composition is a is called b b equals c then indirectly of course at least they call C or something so that it becomes a it can be considered as a category there is many many kind of category but if the category is given then uh the the algebra refracting this categorical structure it is necessary non-commutative uh if that there is a non-trivial arrow between two

options so probability is from the mathematical point of view it's kind of uh the two aspects of the same thing like the fundamental observe of the dependence or contextual dependence or something and this contextuality like a top in top of theory the actually this choosing Concepts and the transformation between the context can be uh formulated in terms of topos which is a very powerful uh kind of category so sorry for the too long explanation but uh without that this very fundamental question it cannot be organized sorry for the long long explanation oh it's it's awesome I mean empirically if you have a sensor that's getting measurements at a certain time scale then there's a binding of the time scale of the of the Observer and the processing um can I ask a question from the chat all right Dave asks at one time Professor tanini was representing the shape of the relations between complexity and Consciousness such that two richly connected systems are non-conscious they're overwhelmed by the complexity recently the inverted U that shows this relation seems to have become scarce leading to undue confusion have IIT people abandoned the idea that there is a bounded sweet zone for Consciousness can a system be either too simple or too complex to act consciously so is IIT lead us to a kind of monotonic Consciousness concept or could there be a U-shaped relationship with information integration and qualia I see yeah I can all see the comment right there so the Indian the question is can the system be either too simple or too complex to act consciously right in the sense yeah from an IIT point of view uh there are several things that uh needs to be said uh but I inverted you is still there in a sense and uh here this person seems to be confusing between the axis of that kind of complexity and the axis of the connectedness you know in IIT original formulation and it's still the true for the IIT 4.0 which is the latest one uh when the system is too connected let's say if you have uh you know 100 neurons and each of them are connected to everybody and in a uniform way then uh uh IIT predicts that you know there it doesn't have much you know uh uh complexity or compositionality or you know uniqueness to arise from that kind of you know system and uh you know even if you cut the system um it doesn't matter much and so that would lead to uh most likely um very low level of consciousness I think that's what yes I should have typed connectivity yes yeah if that's the case that's fine but higher complexity which is granted by some small world or some sparse connectivity structure higher complexity is monotonically associated with more consciousness it's not the case in IIT and that is true from the beginning and the uh easiest kind of you know example is to have a two highly complex kind of system that is uh Loosely connected between them you know it's a kind of a situation where you are interested right in a sense like and Colony yeah each Aunt has a huge brain you know even though it looks extremely small the

number of the neurons they have is more than one medium right I think in that order yes around a million yeah so I think naturally predicted you know the causal connection or um integrated information arising from this one million neurons within my ants are going to be really massive it's very difficult to disconnect in any way uh in uh in your uh meaning of a full way however interaction between a single and to the another has a much less direct causal relationship and therefore the local Minima or local Maxima of the five value is going to be very huge within each and but when you combine or when you treat all these lots of ants it looks it definitely does you know bring some kind of intelligent and complex kind of you know Behavior but phenomenality is not there at that level phenomenality Consciousness within this in a maximum level of you know integrated information like pretty much like you know what we talked about in time scale stuff you know we have a huge you know complex and you know very intelligent kind of behavior and artifacts in the long time scale and there's a short time scale but what matters for our phenomenology is this particular several orders of 100 milliseconds that corresponds to the maximum of the integrated information so you still have emergent intelligence at the colony level but that doesn't necessarily mean that you have qualia arising at that level for analogous reasons why our qualia are at the spiking time scale despite that being only one of the nested time scales exactly exactly intelligence and Consciousness qualia is completely uh you know dissociable and that you know leads to you know the commentary about the large language model probably right is probably structurally built to be very simple most almost like probably you know cerebellum in the human brain which accounts roughly like 80 percent of our neurons in the brain you know out of uh 10 to the 11 neurons in the brain you know eighty percent are in the cerebral but even if you you know remove it or you know lose it or born without it it seems you know Consciousness is not that different and that's uh and a simple kind of prediction that you know cerebellum is doing maybe something similar to the you know large language model is doing it's very modular it's parallel it's not integrated it does a lot of intelligent things but nothing to do with very interesting well maybe in our closing sections um of course anything you want to share or ask but how does active inference benefit and grow from what you're sharing and how do you see things unfolding for people interested in these topics do you want to say something first yeah I was thinking um okay then uh from my and um so I used to um have difficulty in uh you know thinking about the uh active inference or Monica blanket and things like that but you know by broadening my perspective and so becoming more generous I I started to feel like more uh maybe some aspects of the thinking and also former formulation can be yeah relevant at least

you know for the explanation of the biological systems in a various domain and if uh it's uh also you know useful to explain intelligence or construct intelligence that would be and uh from our work of the quantum quality of hypothesis if a quarter is something that is influenced by its measurement then that aspect maybe you know emphasizing this uh usually you know cordial Consciousness is kind of considered a passive you know a construct like at the given moment what I'm experiencing starting passive but now I'm more thinking about because of this you know collaboration with the heart and other people that can't be the case and it's more to do with the process that's a process it may not be that kind of the level of the active inference that you're thinking like you know not the level of the acting on the world but more to do with that this state itself influencing itself might be some kind of you know important uh insight to understand the quality okay I I my comment is the uh okay so the it is uh related to the concept of States again because uh expectation value I I pointed out the importance of expectation value but uh for example what is the expectation value for the for example the cause strength of causality or something then the problem of time scale is very important so the you can uh course grain the for example the brain activity for the one time scales another time scale so I think if IID is correct there is a good time scale which is important of importance regulating importance of the choosing the uh this uh structure then I think the brain information or something called causal structure should be the good category and you can construct a non-commutative algebra and State is defined or something so the time scale is quite fundamental and on the other hand the state on the category as that is a state on the category algebra can be uh uh represented by some kind of weight on arrows the positive kind of weight on arrows so the changing the state is a change in the weight of these hours the positive definite functions I think this the transition of the weighting of the uh this arrow is very uh from at least mathematically very interesting to compare the active inference uh as a mathematical point of view so good uh wait wait to the better weight or something is very hard to say the uh interesting to compare I I'm not uh I don't have much knowledge about active inference but it seems to me it is quite useful to connect this level the state change or something so yeah it is very uh interesting for me and other other thing is I forgot the south point but yeah yeah and yes uh as now already uh okay points out the I I talked about the measurements instruments but the point of the this uh notion is a probe system so measurement is always uh often considered as a very uh direct kind of interaction like the system and the environment but environment has some kind of rope into the system then the interaction is occurred with this Pro Band this system I think it is very analogous to the quoria

the case of so of course the outer world is relating to the brain inside of course at least process over the historically or developmentally but I think the this power through perception was something the brain has its probe over the world for example this interaction is very hard to say in interesting so measurement the analogy to the measurement Theory may be that some physically within the brain but some probe of the outer system under this Korea is interacting so I think the it unifies the picture I think maybe what do you think now for example yeah yeah so that's actually very much uh related to what I was actually constructing this morning yeah so I was reading uh this Ozawa as a paper that describes the instrument and I I now I'm getting better understanding what it means but what he says is that every observable a of a system s is identified with the observable a tensored with the instrument I of a system S Plus S Prime so s here is the original system and S Prime is the one that with any system s uh x s Prime external to s and what the height was meaning is that this S Prime is a probe system which is a part of the environment right yeah and so here what this uh you know Quantum uh initials system description but it's also true for the quantum combination and in my case it is also true for the quantum Courier hypothesis is that uh some kind of query experiencing system is interacting with the word but through the sort of membrane or sort of the like Mark of bracket like thing which is already filtering this is a large interaction of other sensory input and attention and this part itself is kind of you know interacting with itself yeah these arrows are uh the critical part and then if we translate that you know otherwise our way to in our case this could cause not every coil a of a core experiencing system s which is uh in the case of IIT it's called a complex Observer a times instrument over system S Plus S Prime and this S Prime here I think will include all the neuronal processes that support the underlying yeah but it doesn't really necessarily directly in the external environment oh yeah you know many people start to do that but you know if you do that then suddenly you know hallucination or dreaming and things like that becomes very difficult to expand yeah yeah it's including too much yeah I think that corresponds to what you wanted to say right I guess exactly exactly well this is really awesome I very much appreciate that we could have this conversation is there anything else you want to add or ask yeah thank you I also enjoyed and uh it was also good for me to formulate my thoughts as well yeah I'm very surprised uh yeah surprised me in the how consistent uh our thoughts are so how to say that it's it was just a surprise for me um maybe uh the conversation often goes how to say diverse but our conversation has a very surprisingly the conversion so I think it's quite interesting in itself and yeah I run I learned much from this conversation and thank you for inviting us thank you so much so hope to uh join on your

channel and hear updates on your research as things develop so thank you again thank you
thank you bye-bye bye-bye